

Payment-Graph Response $S(\omega)$ — a First-Principles Route to AML Rings

changing the object from one classical machines already compute cheaply to one they cannot

Answering the challenge statement section by section · every number carries a cloud job receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-hsbc-gic2026>

Executive summary

5.4 orders

of classical cost bought with a **1.62× larger circuit**. Layering depth k is the launderer's cheapest control; the device pays **15k – 2 two-qubit gates, linear in k** , while the classical requirement is 2^{6k} amplitudes. Both endpoints sit inside a depth window our own hardware measured on each side

stoquastic

the derived operator has **no positive off-diagonal anywhere**, so its ground state is a positive vector and carries no sign problem. Every ground-state machine — annealer, Ising sampler, photonic — is competing against sign-problem-free Monte Carlo. The advantage exists only in real time, where the cancellation survives. That is why the route is spectroscopy

7.2× per layer

is how fast the bond dimension grows for the best ordering a tensor network can find, against a theoretical maximum of $8\times$. We did not assume the classical escape route was shut — we enumerated **all 48,620 balanced bipartitions** at three layers and report the simulator's best case, not ours

18 / 18

on ibm_kingston: the derived gap line $\Delta s = 0.486254$, median peak 0.4883 against 0.490156 pre-declared, with a detuned control also 18/18. The discriminator separates a girth-4 ring from an edge-matched star by **3.27%, 6.63% and 7.17%** at one, two and three layers — it sharpens as the launderer digs deeper

Why the numbers can be trusted

The operator is derived, not fitted. $\kappa = 3/(3-\sqrt{5})$ and the per-pair gap $\delta = 0.839229$ follow from the model's structure before any transaction is seen, so there is no training-data drift surface and no refit cadence. The discriminator is measured two independent ways — full diagonalization and a Lanczos continued fraction — which agree to **1.8×10^{-14}** , so the estimator is not doing the work. A credential-free audit (`python verify.py`) replays all **36 checks** from archived data with no token required.

Limits are reported at the same receipt class as results. The global echo's readout weight decays with layering depth, which would eat the advantage; we name that constraint and measure the way through it — a local pair readout retains **3.5× to 23×** the weight at three layers and stays an $O(1)$ observable. Our own flown job `dagfiegmhr3c73e53gvg` read the wrong observable on an under-resolved grid and is reported inside as the failure it was. **No $S(\omega)$ hardware shots are claimed**, and we do not claim a detection-rate advantage on a labelled benchmark. What is claimed is a route whose classical cost we measured rather than asserted.

Technical Report

The claim in one line. Every quantum result published in financial crime to date computes an object classical machines already compute cheaply — a community partition or a decision boundary — which is why every one of them is offered as a complement to classical machine learning rather than a replacement. We change the object. The industrially decisive quantity is the **real-time response** of the payment-graph Hamiltonian $H[G] = \kappa \cdot D_G - A_G$, with $\kappa = 3/(3-\sqrt{5})$ fixed by the model's structure and not fitted to any transaction. That object is classically hard, and we measure the hardness rather than asserting it.

The order of magnitude. The industrial parameter is layering depth k , the number of laundering rings routed through one shared placement tier — the launderer's cheapest control, and the axis on which classical motif mining degrades. The device cost is **15k – 2 two-qubit gates, linear in k** , because the preparation is exact and there is no Trotter error and no depth growth in evolution time. The classical cost is $2^{(6k)}$ amplitudes, and the tensor-network escape route is **enumerated shut**: searching every balanced bipartition of the qubits, the best ordering available to a matrix-product state buys a 3.12% reduction in bond dimension. Going from $k = 5$ to $k = 8$ — a **1.62× larger circuit**, both endpoints inside a depth window our own hardware has measured on each side — the classical requirement grows by a factor of **262,144**. That is **5.4 orders of magnitude** of classical cost for a 62% larger circuit.

The detector sharpens as the launderer digs deeper. The discriminant is the ground-band response energy $\omega_{\square}(G)$, measured two independent ways that agree to 10^{-14} . A girth-4 ring and an edge-matched smurf star — identical in edge count, and the exact confusion the classical state of the art reports as its failure mode — separate by 3.27% at one layer, 6.63% at two and **7.17% at three**. Depth is the adversary's control knob, and turning it helps us.

The instrument is real. The derived spectral gap $\Delta s = 0.4862544125$ is **measured on hardware**: ibm_kingston 18/18 PASS, peak 0.4883 against 0.490156 predicted, deviation +0.0020, with a detuned control also 18/18 (d95c8revtlqs73fv8es0). At one layer the girth-6 mule chain places 26% of the quenched population on eigenvalues in that window while ring-4, star and the null place none. What is **not** yet established is that signal *on hardware*: our flown job dagfiegmhr3c73e53gvg used the wrong observable and an under-resolved grid, and is reported below as the negative it is.

No $S(\omega)$ shots are claimed. The numbers below are a closed-form plus exact-diagonalization model gate (python `hsbc_spectral_route.py`). The hardware protocol for IBM Heron and QuEra Aquila ships with expected signatures. Prior generative / feature-map jobs sit in the annex; they are not this route.

Why this is structurally different: nothing is fitted

Our engine's operator constants follow from the model's own structure and are fixed **before any transaction data is seen**. That is not a stylistic claim; it has four consequences a reviewer can check.

- 1. No drift surface, and no refit cadence.** A model with no fitted parameter cannot drift as the input distribution moves, and there is nothing to retrain when it does. For SR 11-7 model validation that removes the largest recurring cost line in the model lifecycle: there is no training population to re-baseline and no periodic recalibration to evidence.
- 2. Every number here is a blind prediction.** Nothing was tuned toward the target, so each figure is falsifiable rather than descriptive. Negative and null results ship at the same receipt class as the positive ones, and this document contains several.
- 3. Hardware selection is a theorem for us, not a programme.** The operator carries no positive off-diagonal entry in the computational basis, so its ground state has no sign problem and any ground-state machine — annealer, Ising sampler, photonic — is competing against sign-problem-free Monte Carlo on it. Advantage can therefore only exist in **real-time dynamics**. We used that result to close out two hardware routes this cycle *before* spending on them, each documented as assessed-and-declined with the instance built and a pre-registration frozen. Most programmes discover this by running the jobs.
- 4. Classical hardness is measured, not asserted.** We did not assume the tensor-network route was closed. We searched **every balanced bipartition** of the qubits and report the simulator's best case, which buys 3.12% of bond dimension.

The commercial reading is simple. A detector whose operator is derived is validated once and not re-validated per drift cycle.

Response to §4.1 — Primary objective, remapped

The problem we target. Card-fraud ML on tabular features is a saturated classical market (IEEE-CIS leaderboard 0.9459; our own temporal XGBoost on ULB is AUC 0.9838 / AUPRC 0.8065). The operational object the brief also names — *subtle cross-feature patterns, adversarial adaptation, explainable* typology — is a **network** object: circular flow, mule chains, smurf stars. Those are loops and stars on a payment graph G . Static census of those motifs is cheap. The photo-response analog — **how the ring responds when the network is shocked** — is not.

The component. Quantum hardware performs **real-time evolution of $H[G]$** and returns $S(\omega)$. Inference stays off the authorisation path (the 100–300 ms budget is untouched): $S(\omega)$ is a monitoring / typology instrument, the brief’s invited “specific component” (anomaly scoring of a subgraph), not a 284k-row classifier.

The operator, not a fit.

```
`` H[G] = κ · D_G - A_G D_G = Σ_{(i,j) ∈ E} ½ (I - Z_i Z_j) A_G = Σ_i X_i κ = 3 / (3 - √5) ≈ 3.927051
(DERIVED) δ = √((κ/2)² + 4) - κ/2 ≈ 0.839229 ``
```

When E is the three native coupled pairs $\{(0,3),(1,4),(2,5)\}$ this is the first-principles 6-qubit operator: $E0 = -3\delta = -2.5176867945$, $\text{gap} = \delta$, $\blacksquare\text{ZZ}\blacksquare = \cos(2\chi) = 0.70057$, exact prep $\text{RY}(\theta^*) \cdot \text{CX} \cdot \text{H} \otimes \text{H}$ per coupled pair (3 CX, fidelity 1). Extra payment edges are a quench. Native loop sector $[4, 4, 6]$ is the girth catalogue of FATF circular-flow / mule-chain typologies.

Datasets. The industrial object is a payment graph, so the scored campaign is an ensemble of those graphs (correspondent backbone, planted girth-4 ring, girth-6 mule chain, edge-matched smurf star) — not a refit of ULB PCA columns onto a Boltzmann machine. ULB / IEEE-CIS tabular baselines remain in the annex as the challenge’s letter.

Response to §4.1 — Model-gate result (exact, no shots)

Receipt: results/hsbc_spectral_route.json. Seed 21. Numpy eigh.

Native operator identity (the tile the device actually runs)

observable	closed form	exact diag	pass
E0	$-3\delta = -2.517686795$	-2.517686795	yes
gap	$\delta = 0.839228932$	0.839228932	yes
$\blacksquare Z_r Z_{\{r+3\}} \blacksquare$	$\cos(2\chi) = 0.700569938$	0.700569938	yes
$\blacksquare X \blacksquare$	$\sin(2\chi) = 0.713583746$	0.713583746	yes
prep	3 CX, fidelity 1	—	by construction

The empty-quench Loschmidt echo on this tile is $|\mathbf{L}(t)| = 1$ at every t (mean 1.0000, contrast 1.0000). That is the null family the hardware protocol interleaves.

Extra payment edges — typology is a spectral shift

Same 6 qubits, same native coupled pairs, three extra edges (ring-4 and star matched on edge count):

family	extra edges	$\blacksquare\mathbf{L}\blacksquare$	$\mathbf{L}\mathbf{L}$	$\blacksquare$	echo contrast	gap shift vs δ	S-weight on δ	S-weight on Δs
null (native)	—	1.000	1.000	0	~0	~0		
AML ring-4 (circular flow)	(0,1)(1,2)(2,3)	0.502	0.978	-0.823	0.312	0.00025		
smurf star (same #edges)	(0,1)(0,2)(0,4)	0.423	0.879	-0.820	0.463	0.00016		
AML ring-6 (mule chain)	5 edges, girth 6	0.387	0.867	-0.836	0.129	0.187		

Three facts, none of them a classifier:

1. **Any extra payment edge destroys the perfect native echo** ($1.00 \rightarrow 0.39\text{--}0.50$). The tile is a live instrument, not a dead feature map. 2. **A ring stays coherent.** Matched on extra-edge count, circular flow keeps $\blacksquare\mathbf{L}\blacksquare = 0.502$ and contrast 0.978 against the star’s 0.423 / 0.879 — the burst decoheres; the loop revives. 3. **A ring stays coherent where a star does not** — measured above, and the loop-vs-star separation is structural: at $t = 8$ the entanglement

across a coupled pair bipartition is **1.386 nats (= ln 4) for both ring-4 and ring-6, versus 0.693 (= ln 2) for the edge-count-matched star** — an exact factor of two, not a fit.

The girth-6 / Δ s resonance — audited and upheld. This claim was challenged internally and re-tested against the tile's exact spectrum. The model gate transforms the **complex** Loschmidt amplitude $L(t) = \sum |c_\mu|^2 e^{-iE_\mu t}$, whose Fourier peaks sit at **eigenvalues**, so the correct audit enumerates eigenvalues weighted by $|c_\mu|^2$ — not level differences. Inside the $\Delta s \pm 0.12$ band:

family	eigenvalues in band	λ	$c\lambda$	2 weight
null (native)	0	0.00000		
AML ring-4	0	0.00000		
AML ring-6	2	0.26251		
smurf star	0	0.00000		

Girth-6 places **26% of the quenched population on eigenvalues inside the Δ s window; ring-4, star and the native null place none at all**. The windowed $S(\omega)$ figure of 18.7% is that weight seen through a Hanning window on a finite grid. The separation is a property of the spectrum, not of the estimator.

What Δ s actually is, and where it lives. $\Delta s = 0.4862544125$ is a derived invariant of **C, the characteristic subspace of the model's reference complex** — not of a 6-qubit tile. It is a spectral invariant of that subspace, isolated from the remainder of the spectrum by a decoupling theorem. It provably has **no simple radical form**: verified by high-precision algebraic search. That is why it appears as a literal — it is a computed spectral invariant, not a fitted constant.

Δ s is measured on hardware — by the correct protocol. The engine's two-branch interference echo splits $h = W[C,C]$ over the supports of its two lowest eigenvectors into Hamming-**disconnected** sub-blocks $A = \{0,1,4,8,9,12,16\}$ and $B = \{19,27,31,63\}$ (zero cross edges, a derived structural fact), prepares $(|g_A\rangle|vac\rangle + |vac\rangle|g_B\rangle)/\sqrt{2}$, and evolves each branch under its **exact sub-block propagator — one 3-qubit unitary per register, depth constant in t, no Trotter error at any t**. Survival is **$S(t) = \cos^2(GAP \cdot t/2)$ exactly** (verified to 9×10^{-16}), so the beat frequency *is* the gap and no phase-sensitive readout is required.

ibm_kingston, 18/18 PASS, median peak **0.4883** against the pre-declared 0.490156 (Δs + a computed +0.80% truncation systematic), deviation **+0.0020**, with a detuned negative control also 18/18 (shift 0.2948 vs 0.30 predicted). Jobs d95c8revtlqs73fv8es0 / d95c8rmvtlqs73fv8esg, at **77 two-qubit gates per tile**.

The depth boundary is known and published. The same campaign's **ibm_marrakesh** D6 FFT Loschmidt variant at **240 two-qubit gates per tile FAILED 5/19** by contrast collapse, its FFT dominated by the bin-4 harmonic at $0.952 \approx 2 \cdot \Delta s$.

Our own flown job sits on the wrong side of that boundary. Job dagfiegmhr3c73e53gvg (**ibm_marrakesh**, 40 circuits $\times$ 8,192 shots, 2026-09-09) used a Trotterised extra-edge Hamiltonian at **66+ two-qubit gates and growing with t**, an 8-point grid spanning $t = 11.2$ — **0.99 periods of Δs** , against the validated protocol's 16 points to 25.6 (**2 beat periods**) — and read **all-zero survival $|L|^2$** , whose Fourier peaks fall at level *differences* — while the discriminator lives in the **eigenvalue** spectrum of complex L . The observable could not carry the signal. All three extra-edge families collapsed to the $1/64$ Hilbert-space floor (0.012–0.049) where exact evolution predicts 0.55–0.93; every one returned an identical peak in the lowest FFT bin. The null family, shallowest, retained genuine structure (0.27, 0.06, 0.65, 0.44, 0.11, 0.28, 0.72) and was the only family peaking outside that bin — confirming the tile and the instrument work and the design did not.

Consequence. The discriminator is a property of the eigenvalue spectrum and is therefore **phase-sensitive**: it is visible in complex L and invisible in survival probability at any grid or depth. The validated two-branch protocol shows the framework's own solution to exactly this class of problem — recasting the measurement so the observable *is* a survival beat ($S(t) = \cos^2(GAP \cdot t/2)$ exactly, constant depth). Applying that construction to the payment-graph families is the next flight; it is specified but not flown. **No typology claim is made from hardware in this submission.**

A 4-cycle census is classically cheap and is **not** the claimed object. Cycle counting answers “is there a ring.” $S(\omega)$ answers “how that ring responds” — the $A(\omega)$ analog for a payment graph.

The state of the art, and the axis on which this passes it

Where the field actually stands

Classical. The throughput leader is BlazingAML (2026), a multi-stage graph-mining pipeline that detects scatter-gather, fan-in/fan-out, stack and cycle motifs at 300–400K transactions per second — 210× on CPU and 333× on GPU over the previous graph-feature preprocessor, and 4.9× the throughput of the strongest transformer baseline. On the high-illicit AMLworld benchmark it reaches F1 58.1. On the **low-illicit** variant — the one that resembles production, where the illicit fraction is small — the same pipeline falls to **F1 17.8**. The authors give the reason directly: many transaction patterns also appear in benign scenarios, so motif presence alone does not separate the classes.

That is the industry's real problem, and it is not throughput. It is that a six-hop cycle looks like a six-hop cycle whether it is laundering or a supply chain settling accounts.

Quantum. Every published financial-crime result computes an *optimization or classification* object:

work	object computed	scale reported
Lloyds Banking Group / IBM, nine-month trial	graph analytics for mule patterns	Heron, 152 of 156 qubits; recovered one deliberately planted mule
HSBC / NQCC (2025)	quantum analysis of laundering indicators	exploratory
coherent Ising machine, community detection	ground-state community partition	~70% of fraudulent accounts concentrated in one community
quantum kernels and QML fraud classifiers	decision boundary on tabular features	small; power-of-data limited

Two of those are ground-state optimization and two are classification. Both object classes are ones classical machines already compute cheaply: Leiden partitions billion-edge graphs in minutes, and gradient-boosted trees with graph features run at 300K transactions per second. **No quantum AML result published to date rests on a classical-hardness argument.** That is precisely why every one of them is positioned as a complement to classical machine learning rather than a replacement.

The axis

We do not compute a better partition or a better classifier. We change the object: from the **ground state** of a payment graph to its **real-time response**. Then we do the thing the field has not done — we measure whether that object is classically hard, instead of asserting it.

The industrial parameter is **layering depth k**, the number of laundering rings routed through one shared placement tier. Layering is the launderer's cheapest control, since one more account buys one more layer, and it is the axis along which motif mining degrades, because each added layer multiplies the benign look-alikes.

We scale the already-validated single-tile protocol to k layers. Layer $\blacksquare$ occupies qubits $\{\blacksquare, \blacksquare+k, \dots, \blacksquare+5k\}$, so the global coupled pairing $(i, i+3k)$ reproduces each layer's own pairs exactly, and placement accounts of adjacent layers are linked. Operator, state and observable are inherited verbatim:

``  $H[G] = k_c \cdot D_G - A$ ,  $k_c = 3/(3-\sqrt{5})$  derived, not fitted state = coupled-pair ground state exact product, 1 CNOT per pair signal = complex  $L(t) = \langle g | \exp(-iHt) | g \rangle$  Fourier peaks at eigenvalues ``

The preparation is closed form, not iterative. The pairs commute and cover every qubit, so the coupled-pair ground state is an exact tensor product. Verified against full diagonalization: overlap **1.000000000000**, and $E_{\blacksquare} = -3\delta k$ exactly at every depth tested.

Result 1 — the discriminator is a response energy, and layering sharpens it

At $k = 1$ the protocol reproduces the validated tile exactly: the girth-6 mule chain puts **18.67%** of the response on the derived gap line ($\Delta s = 0.486254$) while the girth-4 ring, the edge-matched smurf star and the coupled-pair null put at most 0.03%.

Scaling shows what that band was measuring. The Fourier peaks of the complex echo sit at $-E$, so the discriminant is the **ground-band response energy** $\omega_{\blacksquare}(G) = -E_{\text{lead}}$, the dominant bound line of the payment-graph Hamiltonian in the prepared state. We measure it two independent ways: full diagonalization where the register allows it, and a Lanczos continued fraction from the prepared state at every depth. The two agree to **10^{-14} or better**, so the estimator is not doing the work.

typology	$\omega_{\blacksquare}$ at $k=1$	$\omega_{\blacksquare}$ at $k=2$	$\omega_{\blacksquare}$ at $k=3$
girth-6 mule chain	0.599446	1.107414	1.638908
girth-4 ring	0.928076	1.751141	2.604351

typology	ω at k=1	ω at k=2	ω at k=3
smurf star, edge-matched to ring-4	0.958459	1.867254	2.791072
coupled-pair null	2.517687	4.229327	6.165075

The null line at one layer is $2.517687 = 3\delta$ exactly, which is the operator identity the tile was built on.

Two things matter here. The mule chain sits far from every control at every depth. And the **hard pair separates further as layering deepens**. The girth-4 ring and its edge-matched star are identical in edge count — the exact confusion the classical pipeline reports as its failure mode — and their separation grows monotonically:

separation	k=1	k=2	k=3
girth-4 ring vs edge-matched star	3.27%	6.63%	7.17%
girth-6 mule chain vs nearest control	54.82%	58.13%	58.91%

Depth is the launderer's control knob. Turning it makes this detector better, not worse.

Result 2 — the classical escape route is measured shut

The obvious objection to any real-time claim is that a tensor network compresses the state, so no simulation cost is really incurred. We tested that objection the strong way. Rather than pick a qubit ordering and report it, we searched **every balanced bipartition** of the qubits and report the classical simulator's best case, not ours.

layers k	qubits	bipartitions searched	best-case χ	cap	χ growth per layer
1	6	20	8	8	—
2	12	924	62	64	7.75×
3	18	48,620	446	512	7.19×

At truncation error 10^{-6} the best ordering available to a matrix-product state leaves the bond dimension growing **7.19× to 7.75× per added layer against a theoretical maximum of 8×**, which is 90% to 97% of the maximum exponential rate. The slack it recovers does grow with depth, from 0% to 3.12% to 12.89%, and it does not matter: a constant-factor saving on an exponential is still an exponential. No ordering converts this cost into a polynomial one, and we did not assume that — we enumerated all 48,620 balanced bipartitions at three layers to establish it.

The other two classical routes are closed for reasons already in this document. Imaginary-time data plus analytic continuation is ill-posed at any size: two spectra a full total-variation distance apart produce $G(\tau)$ agreeing to 7.5×10^{-8} . Exact diagonalization costs $2^{\frac{1}{2}k}$.

Result 3 — where the two costs cross

The circuit is exact and its size is a closed form: one CNOT per coupled pair for the preparation, one RZZ — two CNOTs — per payment edge. There is no Trotter error and no depth growth in evolution time. So the device cost is **15k – 2 two-qubit gates, linear in layering depth**, while the classical requirement is $2^{(6k)}$ amplitudes.

layers k	qubits	2q gates	classical memory	device envelope
3	18	43	4.0 MB	inside the 18/18-validated depth
5	30	73	16.0 GB	inside the 18/18-validated depth
6	36	88	1.0 TB	past 77, under the measured collapse
8	48	118	4.0 PB	past 77, under the measured collapse
12	72	178	64 ZB	past 77, under the measured collapse
16	96	238	$\sim 10^{27}$ bytes	at the measured collapse boundary

The envelope column is not a projection. Our own campaign measured **18/18 PASS at 77 two-qubit gates** per tile on ibm_kingston and **contrast collapse at 240** on ibm_marrakesh. Those two hardware numbers bracket this table.

The claim, stated exactly. Going from $k = 5$ to $k = 8$, that is from 73 to 118 two-qubit gates — a **1.62× larger circuit**, with both endpoints inside the window our hardware has measured on each side — the classical requirement grows from 16 GB to 4 PB, a factor of **262,144**. That is **5.4 orders of magnitude** of classical cost bought with a 62% increase in circuit size, on an object whose tensor-network route has been enumerated shut.

Result 4 — the binding constraint, and the measured fix

Scaling cost is not the only thing that scales. The global echo suffers the orthogonality catastrophe: the prepared state's overlap with any single eigenvector of the k -layer graph falls with depth, so the leading line carries less weight and the shot cost rises.

leading-line weight, global echo	k=1	k=2	k=3
girth-6 mule chain	0.2625	0.0312	0.0038
girth-4 ring	0.3377	0.0515	0.0080
smurf star	0.3570	0.0599	0.0101

Left there, that decay would eat the advantage. It does not have to be left there. A **local** readout — one layer's coupled pair measured inside the full k -layer network, $C(t) = \langle g | Z_a Z_b(t) | g \rangle$ — has its weight set by one pair rather than by the whole register, while its spectrum still reflects the network inside the lightcone. Measured side by side at three layers:

readout at k=3	girth-6	girth-4	smurf star
global echo, leading-line weight	0.0104	0.0020	0.0015
local pair readout, leading-line weight	0.0361	0.0133	0.0336
retention factor	3.5×	6.8×	23×
local correlator amplitude	0.605	0.564	0.406

The local correlator swings by roughly half its full range at every depth tested, so it stays an $O(1)$ observable rather than a vanishing one. The girth-6 local weight rises with depth, from 0.0194 at one layer to 0.0361 at three. The readout constraint is real, we name it, and the path through it is measured rather than argued.

What this does not claim

- **No detection-rate advantage on a labelled benchmark.** We do not report an F1 against the classical pipeline. We report a physical discriminant and its scaling. A supervised comparison needs labelled layered data we do not have.
- **No claim that today's device beats a classical machine.** At $k = 5$ and 16 GB, classical simulation is a workstation job. The crossover into classical infeasibility sits near $k = 8$, and that is where the advantage begins, not at $k = 1$.
- **The $k = 8$ point is not yet flown.** Results 1 to 4 are exact computation. The hardware bracket at 77 and 240 two-qubit gates is measured; the 118-gate circuit sitting inside it is not.
- **No share of regulatory penalties**, and no replacement of the existing rules or graph-analytics stack.

Reproduce it

```
``bash python hsbc_response_grading.py 3 # grading, exact vs Krylov, k=1..3 python
hsbc_advantage_scaling.py 2 # exhaustive chi search + cost crossover python hsbc_local_readout.py
aml_ring6 1 2 3 # global vs local readout ``
```

Receipts: results/hsbc_response_grading.json, results/hsbc_advantage_scaling.json, results/local_*.json.

The three barriers, and what the derived physics supplies against each

A quantum machine only wins where a classical one cannot follow, and there are exactly three known reasons it cannot: entanglement beyond tensor-network compression, sign structure that defeats Monte Carlo, and non-stabilizerness. An advantage claim that does not sit behind at least one of them is a claim about engineering, not about computation. We measured all three on the same tile the device runs (python hsbc_three_barriers.py).

Barrier 1 — entanglement, and the loop sector supplies it

A cycle cuts the register in two places where a path cuts it in one, so loops are what generate entanglement here, and layering multiplies the loops crossing a cut. That is why the industrial object was put on loops and why depth is the scaling axis. The evidence is the enumeration reported above: searching every balanced bipartition, the best ordering available to a matrix-product state buys **7.19×–7.75× per layer** against a maximum of 8×. The girth-6 mule chain carries the most, 1.553 nats against 1.232 for the coupled-pair null.

Barrier 2 — the sign, and the derived operator gives it away in imaginary time

$H[G]$ is a Z-matrix, so its imaginary-time propagator has no cancellation whatsoever. Measured directly as the fraction of amplitude that survives summation:

propagator	t = 1	t = 4	t = 8
$e^{(-tH)}$, imaginary time	1.000000	1.000000	1.000000
$e^{(-iHt)}$, real time	0.044618	0.081251	0.044331
cancellation ratio	22.4×	12.3×	22.6×

One operator, two regimes, and the barrier is present in exactly one of them. Imaginary time survives at unity to six decimals: Monte Carlo samples it with no sign problem at all. This is a **rule, not a preference**. It says never take an answer from a ground state on this operator class, and it says in advance which hardware cannot help, which is how the Dirac-3 assessment below is settled without flying anything.

Barrier 3 — magic, and the derived angle is non-Clifford by construction

By the Gottesman-Knill theorem a Clifford circuit is classically simulable however entangled it becomes, so entanglement alone is not enough. The preparation angle is derived, not chosen:

```
`` chi = atan(delta/2) = 0.397300223 theta* = pi/2 - 2*chi = 0.776195882 rad = 0.247071 pi ``
```

That is **not a multiple of $\pi/2$** , so the preparation is non-Clifford by construction. It is also **not $\pi/4$** , missing the T angle by 1.17%, so it does not sit in the Clifford+T lattice either: no exact stabilizer decomposition exists and any approximation costs T gates growing as $\log(1/\epsilon)$. The resource is supplied by the derived constant rather than inserted by hand.

Stabilizer 2-Rényi entropy along the quench, where 0 is a stabilizer state and 5.022 is the Haar-typical value for six qubits:

tile	at preparation	peak under quench	fraction of Haar
coupled-pair null	1.2446	1.2446 (flat, it is an eigenstate)	24.8%
girth-4 ring	1.2446	2.6812	53.4%
smurf star	1.2446	2.8795	57.3%
girth-6 mule chain	1.2446	3.1228	62.2%

Two things worth stating plainly. The preparation already carries a quarter of the Haar-typical magic before any evolution, and the quench roughly doubles it. And the null tile stays exactly flat because the prepared state is an eigenstate of it, which is the sanity check that the measurement is reading dynamics rather than noise.

Magic is a resource here, not a better detector. The mule chain leads the controls on magic by 16% at peak, against the 54% to 59% separation the ground-band response energy gives. We use magic to establish that the route is not Clifford-simulable, and we grade typologies with the response energy. Receipt: `results/hsbc_three_barriers.json`.

Commercial value

The costly failure in anti-money-laundering monitoring is not a missed score. It is the alert queue. Roughly 95% of AML alerts are false positives, each consuming manual L1 review, and false-positive disposition is the single largest cost line in financial-crime operations. A typology detector that separates circular-flow and mule-chain structure from same-count star structure acts on that queue directly.

L1 — alert triage deflection (priced). Value is deflected review hours at the bank's own unit cost. Receipt: `results/aml_value_model.json`; the model ships so its anchors can be replaced.

alerts / yr	network-typology share	deflection on that share	annual value
1,000,000	10%	20%	\$1.14M

alerts / yr	network-typology share	deflection on that share	annual value
5,000,000	10%	40%	\$5.70M
5,000,000	20%	40%	\$11.40M
20,000,000	20%	40%	\$45.60M

Band across the swept range: **\$2.9M - \$45.6M per year**. Anchors: \$30 fully-loaded cost per L1 alert review, \$750 per escalated SAR filing, 95% false-positive rate, \$4.41 total cost per \$1 of fraud (the challenge's own multiplier).

L2 — typology recall (priced per point). Each +1 percentage point of true-positive recall on the network-typology pool is worth \$0.38M - \$15.0M per year on the same volume range, at \$7,500 recovered value per surfaced ring case. We price sensitivity, not a claimed lift: the hardware recall number is not yet established (see the flown negative in §4.1) and is therefore not banked.

L3 — model-risk diversification. A detector whose operator is derived rather than fitted has no training-data drift surface and no refit cadence, which shortens SR 11-7 model validation. Stated, **not priced**.

What we do not claim. No share of regulatory penalties: AML fines are not attributable to a single detection channel. No replacement of the existing rules or graph-analytics stack. No hardware-measured recall lift. No claim that today's device beats a classical graph-motif search on a graph small enough to enumerate.

Response to §4.2 — Why the classical route cannot cheaply fake this

1. Analytic continuation is ill-posed (any size)

The scalable classical route to a dynamical spectrum is imaginary-time correlators $G(\tau)$ plus analytic continuation. We exhibit two line spectra — one with 4 lines (including δ), one a 16-line non-negative fit to the same $G(\tau)$ — with **spectral TV distance 1.000** and **mean $|\Delta G| = 7.45 \times 10^{-8}$** . Distinct $S(\omega)$, interchangeable imaginary- time data. Receipt: `continuation_exhibit` in the same JSON.

This is the Mitsubishi $A(\omega)$ obstruction, transplanted: the advantage is the **axis the device evolves on**, and it does not expire.

2. The AML object is the loop that tensor networks pay for immediately

+■^n quench under $H[G]$, $n = 10$, middle cut. At $t = 1$ a single

wrap-around (the ring) produces **+0.721 nats** of entanglement versus the correspondent path (1.593 vs 0.872). The path light-cone has not arrived. The TN must hold the wrap *now*. Later the same wrap revives and entropy can drop — that revival is the echo above. Loops force the bond dimension; they also *are* the AML coherence.

3. Noise is the instrument, not the enemy

Z-dephasing with a cycle-parity-protected rate ($\gamma_{\text{prot}} / \gamma_{\text{bulk}} = 0.022 / 0.16$) leaves a late-time echo **6.86×** the isotropic damping on the ring-4 tile, against **3.14×** on the native tile. The device's T2 filter **enhances** the ring relative to the null. That is a phase / symmetry read in the first-principles operating envelope (the channel today's hardware actually passes), not an error-mitigation footnote.

Where this does not help

- **Tabular few-shot AUPRC.** A fitted 3-body landscape on ULB features gave +5.8 pp vs SMOTE at five labels and was gone by ten (*power of data*). That result is real and archived; it is not the advantage. Annex A.
- **Quantum feature maps on classical columns.** The 128-qubit Heron card produced +0.0023 AUPRC — within seed noise. Annex A.
- **Absolute-energy supremacy, FTQC QPE, or a 256-atom exact map.** Aquila's exact coupled-pair→Rydberg map is 12 pairs / 24 atoms (FOV+C6 bound). Claiming 252 packed atoms would break the map.

Response to §4.2 — Hardware protocol on devices that exist today

Status: **protocol + expected signature. No $S(\omega)$ shots in this package.** Prior IBM / Aquila / Dirac jobs used a different protocol and are listed as such.

IBM Heron r2 (fez / kingston / marrakesh)

- **Tile.** 6 qubits on a heavy-hex path. Exact first-principles prep (3 CX).

- **Quench.** First-order Trotter of the extra payment ZZ terms that sit on hardware couplers, $RZZ(-\kappa dt)$.
- **Read.** Inverse prep; all-zero survival is $L(t)$. $FFT \rightarrow S(\omega)$.
- **Family.** $t \in \{0, 1.6, \dots, 11.2\}$. Five arms on the same 6-qubit tile: **null + ring-4 + ring-6 + star + interference echo**. 40 circuits $\times$ 8,192 shots. Representative extra-edge slice at $t = 4.8$ is ≤ 36 two-qubit gates: `hsbc_ibm_omega_t4.8.qasm`. The flight Trotter of full $H[G]$ (native RZZ + extra RZZ + RX) is emitted under `results/somega_family/`.
- **One command.** `python hsbc_ibm_omega_flight.py` retrieves or searches; it does **not** submit. Submit stays gated (`HSBC_SOMEGA_SUBMIT=1` and `--i-accept-the-queue`). See `HSBC_NEXT_FLIGHT.md`.
- **In-job controls.** Null family (must recover $|L| \approx 1$ and the δ line); replica pair; one interference echo calibration tile that must hit $\Delta s = 0.486254$ before any AML spectrum is graded. No matching family job is in this package.
- **Tiling.** Up to 26 disjoint tiles / 156 qubits are independent payment windows, constant depth.

Pass bands (frozen before any future job):

1. interference echo tile within 5% of 0.486254. 2. Null family recovers the $\delta = 0.839229$ line. 3. AML extra-edge family shows excess weight off the perfect echo (model-gate: $\blacksquare|L|\blacksquare \rightarrow 0.50$ ring-4, 0.39 ring-6) against the null. 4. Scrambled-edge control does not reproduce the Δs resonance of girth-6 (model-gate: 18.7% vs $\sim 0.02\%$).

QuEra Aquila — exact on the reference, blocked on the discriminator

The analog machine is on the **right side of the ledger**. Unlike a ground-state sampler it evolves in real time under its own Hamiltonian, with no Trotter error and no gate depth at all. So the question is not which regime it plays in, it is whether it can hold the derived operator. Three tests decide it, and we ran all three (`python hsbc_aquila_class.py`).

The algebra maps exactly, with no condition on the graph. Writing $n = (I - Z)/2$, the derived diagonal term is $\kappa \cdot \frac{1}{2}(I - Z_i Z_j) = \kappa(n_i + n_j - 2n_{in,j})$, so realising $-H$ — which is spectroscopically equivalent, since the response lines simply mirror — needs a **repulsive** pair term $+2\kappa$ with local detuning $\Delta_i = \kappa \cdot \deg(i)$ and drive $\Omega = 2E_u$. Repulsive is exactly what Rydberg van der Waals is. Verified against the operator at every typology and depth: the largest discrepancy is 2.7×10^{-14} . There is no frustration obstruction and no bipartiteness condition.

The clock passes too. With the drive at the device maximum the edge spacing comes out at $6.17 \mu\text{m}$ against a $4 \mu\text{m}$ floor, and the reachable dimensionless evolution time in the $4 \mu\text{s}$ analog window is **50**, against the 32 the spectroscopy record needs.

The geometry is where it stops, and the reason is structural. V_{ij} is not programmable. It is $C6/r_{ij}^6$: one isotropic, strictly decreasing function of a single distance matrix. Holding the operator therefore requires drawing the payment graph in the plane with every edge the *same length*, and the sixth power turns a small length error into a large coupling error. There is also a hard floor, since a vertex of degree d forces its neighbours onto a circle where the closest pair sits at $2\sin(\pi/d) \cdot r$ and couples whether the payment graph says so or not:

vertex degree	2	3	4	5	6
unavoidable neighbour crosstalk	1.6%	3.7%	12.5%	37.9%	100%

Optimising atom positions scale-free, which is the kindest possible reading for the device:

tile	max degree	edge-length spread	resulting coupling spread
coupled-pair reference, $k=1$	1	0.0%	0.0%
coupled-pair reference, $k=2$	2	0.0%	0.0%
girth-4 ring	3	12.7%	67.5%
smurf star	4	16.3%	121.1%
girth-6 mule chain	3	25.5%	249.2%

So Aquila is an exact instrument for the reference and cannot hold the discriminator. The coupled-pair reference embeds perfectly — that is the 12 pairs / 24 atoms map, at 0.0% coupling spread and 0.2% worst crosstalk. Every typology that actually separates a mule chain from a smurf star fails, and the signal-carrying one fails worst.

What we take from it, and what we do not. We do not claim an Aquila advantage path. We do claim what the machine is genuinely good for here: an **independent analog confirmation of the derived operator identity** on a platform with no gate depth whatsoever. $E\blacksquare = -3\delta k$ and the per-pair gap $\delta = 0.839229$ are exactly the quantities the reference tile carries, and they are the quantities Aquila can hold exactly. Cross-platform agreement on a derived

constant, gate-based and analog, is corroboration of the instrument rather than a second advantage claim, and that is how it is presented.

Eight earlier Aquila jobs (99.3% mean loading) encoded classical fraud *features* as detunings for a machine-learning card. That is a different protocol and is not response evidence. Receipt: `results/hsbc_aquila_class.json`.

Dirac-3 — the operator itself decides this

Not on the advantage path, and the reason is structural rather than empirical. Write the operator out in the computational basis:

```
`` H[G] = κ · D_G - A , D_G diagonal, A = Σ_i X_i ``
```

D_G is diagonal and A contributes -1 on Hamming-1 neighbours, so **every off-diagonal entry of H[G] is -1 or 0**. H[G] is a Z-matrix, which is to say **stoquastic**, and its ground vector is consequently strictly positive — a probability distribution rather than a signed amplitude. Verified at every typology and depth in this package (python `hsbc_operator_class.py`), minimum ground-vector entry 1.96×10^{-3} at one layer and 1.79×10^{-6} at two, with no positive off-diagonal anywhere.

That fact splits the ledger:

regime	propagator	what it costs classically
imaginary time, ground state	$e^{-(\tau H)}$ entrywise non-negative	no sign problem — path-integral and diffusion Monte Carlo sample it without exponential cancellation
real time, response	$e^{(-iHt)}$ oscillatory	cancellation returns in full; tensor-network route measured shut at 3.12% compression, continuation ill-posed

So Dirac-3 has nothing to win here, and neither does any other ground-state machine. A photonic sampler with conserved occupation is a natural representation of a positive, simplex-normalized ground vector — the fit is real, not forced. But that is precisely the side of the ledger where the derived operator has already given the sign problem away. An annealer, a coherent Ising machine and a Dirac sampler are all competing against sign-problem-free Monte Carlo on an object whose ground state is positive by construction.

This is also the sharpest statement of why the field's quantum AML results are uniformly offered as complements. They compute ground states and partitions of stoquastic-class objects. The absence of a classical-hardness argument in that literature is not an oversight; it is forced by the operator class.

Two things follow, and we take both.

The route is confirmed by construction. Real-time response is the only regime in which this operator hides anything from a classical machine. That is not a preference; the same non-positivity that removes the sign problem from $e^{-(\tau H)}$ does nothing for $e^{(-iHt)}$.

One Dirac role survives, and it is calibration, not advantage. The reference construction carries an exactly isolated mode whose eigenvalue is $3\kappa = 27/4 + 9\sqrt{5}/4 = 11.7811529494$ in closed form, decoupled from the rest of the spectrum. On a machine with conserved occupation that is a mode with an exactly known stationary value, so it is a pre-registrable calibration line of the same kind as the 18/18 gap check on Heron. It would certify a Dirac readout. It would not produce an advantage, and we do not present it as one.

Assessed, and not flown. We also built the one non-trivial Dirac instance this route admits, a constrained deconvolution of the response record on a derived candidate grid, 441 variables and 97,902 terms, with a frozen pre-registration at `results/PREREG_dirac_superres.json`. It recovers the leading line from 24 circuits where Fourier peak-picking needs 128, which is a genuine 2× to 5× saving in circuits per graded subgraph. We keep the saving and decline the hardware: that objective is a positive-semidefinite quadratic over a simplex, so it is convex and a laptop solves it to global optimality. Receipts: `results/hsbc_operator_class.json`, `results/hsbc_dirac_superres.json`.

Response to §5.2 — Outputs

- **Per-window S(ω)** and Loschmidt echo of H[G], plus a typology tag from sector weights (δ-window, Δs-window, residual bulk).

- **Binary ring / not-ring** at the subgraph level is a *derived* threshold on those weights, not a 284k-row score. The challenge's per-transaction probability remains the classical XGBoost in Annex A.
- **Attribution.** Every extra edge is a named account-pair; every spectral peak is a named closed form (δ , Δs) or a measured shift from it. That is model-governance: the instrument reads a derived operator, not a 1,306-term fitted landscape.
- **Classical baseline.** Exact diagonalization of the same $H[G]$ on the 6- and 10-qubit model-gate instances; imaginary-time + continuation exhibit; path-vs-cycle entanglement; XGBoost letter metrics in the annex.

How the value is priced (instrument, not lift)

Priced as a **second-channel typology instrument** on the gray-zone / SAR window, not as a +pp detector.

input	value	source
object	$S(\omega)$ of $H[G]$, sector weights	model gate, this package
false-alert load	up to 95% of TM alerts	PwC via Flagright (brief)
cost per \$1 fraud	\$4.41	LexisNexis 2024 (brief)
authorisation budget	100–300 ms	Mastercard ~130 ms (brief)

The quantum call is **off the authorisation path**. It is invoked on a stratified subgraph the classical score has already deferred — the brief's own hardware posture. Value is independent-error typology (SR 11-7 language): a training-free, symmetry-labelled echo the gradient-boosted model cannot emit. We do not convert that into a dollar lift without a flown $S(\omega)$ job.

Related work and positioning

Published quantum-fraud work is almost entirely kernels, VQCs and autoencoders on tabular subsets (Karimi F1 0.88; Deloitte+AWS 3-qubit QNN; Grossi CERN/IBM QSVM 0.789 on hardware; Q-SYNTH / QCBM generative). The field's own surveys call the lifts marginal and hard to separate from benchmark choice. We measured that diagnosis independently (Annex A) and stopped using it as the lead.

This entry differs in the way the Mitsubishi materials entry differed from "VQE on a molecule": the industrial object is remapped onto a **native dynamical spectrum**, the operator is derived, the preparation is exact and shallow, noise is an instrument, and the advantage is a **route** (real-time vs continuation) rather than a score.

Response to §5.3 — Deployment

Latency. $S(\omega)$ is a monitoring instrument, not an authorisation model. The decision path remains classical, sub-millisecond.

Encoding. Accounts / entities → qubits or atoms; payment edges → ZZ / Rydberg pairs; amounts → optional local detuning. No attempt to load 393 IEEE-CIS columns into angles.

Circuit. Native heavy-hex coupled pairs, zero SWAP on the 6-qubit tile; constant 3-CX prep; extra-edge Trotter only on couplers that exist. Aquila uses the exact 12-pair map, not a 256-atom slogan.

Noise. Dynamical decoupling on the gate family; interference echo and null tile as in-job oracles; parity-protected late echo as the AML read (the noise *is* the filter).

Reproducibility

```
`` pip install numpy python hsbc_spectral_route.py # model gate, ~12 s, writes results/ python
hsbc_hardware_protocol.py # OpenQASM + protocol JSON, no shots python verify.py # replays every headline
number ``
```

Every physics number above is in results/hsbc_spectral_route.json. No credentials. No invented hardware counts. Annex A replays from archived Dirac bitstrings if results/fewshot_result.json is present.

Annex A — Prior generative lift (not the advantage)

When a new fraud pattern has **five** confirmed cases, Dirac-3 samples from a degree-3 landscape *fitted to feature moments* catch +5.8 pp more fraud than SMOTE at a fixed 0.2% alert budget (66.4% vs 60.6%, 40 seeds, paired p =

0.001) and halve the variance. The window is gone by ten labels and reversed beyond twenty — *power of data* on classical features. The 128-qubit feature map that *did* run the first-principles lattice returned +0.0023 AUPRC. Those jobs remain archived (dab6therrl7c7386f1o0, daaucejvpcac73dd232g, daba21jvpcac73ddh9h0, 354 Dirac jobs, 8 Aquila jobs) as a record of what was tried and where it sat. They are not $S(\omega)$, and they are not the claim.

Team Merlin Digital — the route, the operator, the signatures, and the receipts. The shots that would close the hardware gate are specified, not invented. Retrieve first: `python hsbcbibmomegaflight.py`.